

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 6. Loops - Control Flow

Lecture: 56. Challenge - Finding Max and Min Number

Section 1: Lecture Summary

This lecture focuses on solving programming challenges using Python's `while` loop to find the maximum element in a set of user-input numbers, and then extends the challenge to find both maximum and minimum elements simultaneously. The instructor explains the logic step-by-step, starting from outlining program requirements, discussing key variables and conditions, showing the internal logic of comparisons, and finally demonstrating code implementation in Jupyter notebook. The lesson emphasizes maintaining variables to keep track of the current maximum and minimum values, using a counter variable to iterate inputs, input typecasting, and proper initialization of these tracking variables. The challenge encourages learners to write code for finding the maximum element first and then extend it by adding minimum element tracking, using appropriate initial values such as negative and positive infinity to support negative and positive numbers.

Section 2: Key Concepts and Explanations

- Using `while` Loop for Repetitive Input

- A loop is necessary to repeatedly take multiple integer inputs, one at a time.
- A counter variable (`i`) is initialized to zero and incremented till it reaches the number of elements (`n`).

- Tracking Maximum Element

- Initialize a variable `max` to zero (assuming all inputs are positive).

- For each input `x`, check: if $x > \text{max}$, then update $\text{max} = x$.
- This ongoing comparison inside the loop ensures `max` always holds the greatest number so far.

- Taking User Input

- Use `input()` function without prompt inside the loop to take numbers.
- Since `input()` returns a string, convert it using `int()`.

- Program Steps Breakdown

1. Print a message asking the user how many elements to enter.
2. Initialize counter and max variable.
3. Use a `while` loop to take each element input.
4. Update max accordingly inside loop.
5. After loop ends, print `max`.

- Extending to Find Minimum Element

- Declare another variable `min`.
- Update logic to check minimum similarly: `if x < min`, then `min = x`.
- Both max and min comparisons should be separate conditions; not mutually exclusive.

- Initialization for Handling Negative Numbers

- For general cases (mixed positive and negative numbers) initialize variables as:
 - `max = float('-inf')` # Represents negative infinity - smaller than any number.
 - `min = float('inf')` # Represents positive infinity - larger than any number.
- This ensures any entered number replaces these initial extreme values correctly.

- Important Note on Conditional Logic

- Do not use an `else` clause assuming input is minimum if not maximum.
- Example: If `x` is not greater than `max` but neither less than `min`, no update occurs.
- Both conditions must be checked independently each time.

- Alternate Initialization Method

- Initialize `max` and `min` to the first input number before the loop starts.
- Then loop for remaining numbers to update max and min.

Section 3: Example Code and Use Cases

Finding Maximum Element:

```
n = 5
print('Enter', n, 'Numbers')
i = 0
max = 0 # Assumes all positive numbers

while i < n:
    x = int(input())
    if x > max:
        max = x
    i += 1

print('Max Element:', max)
```

Explanation:

The program asks for 5 numbers, reads each using a `while` loop, and keeps updating `max` if the new number is greater than the current `max`. Finally, it prints the largest number entered.

Finding Maximum and Minimum Elements (with initialization to infinities):

```
n = 5
print('Enter', n, 'Numbers')
i = 0
max = float('-inf') # Negative infinity
min = float('inf')  # Positive infinity

while i < n:
    x = int(input())
    if x > max:
        max = x
    if x < min:
        min = x
    i += 1

print('Max Element:', max)
print('Min Element:', min)
```

Explanation:

This program similarly takes 5 numbers but tracks both maximum and minimum values using two variables initialized to extreme float values, ensuring the program works correctly with any integer input values (positive, negative, or zero).

Section 4: Key Takeaways

- A `while` loop with a counter variable lets you process a fixed number of inputs dynamically.
- Maintain a `max` variable to store the current maximum; update it inside the loop by comparing each new input.

- To find minimum alongside maximum, maintain a separate ``min`` variable and independently check for minimum condition.
- Proper initialization of ``max`` and ``min`` is crucial:
 - For only positive inputs, ``max = 0`` can suffice.
 - For mixed inputs, use ``float('-inf')`` for max and ``float('inf')`` for min.
- Always check conditions for max and min separately; do not assume else part equates to the other.
- Using the first input value to initialize both ``max`` and ``min`` is a valid alternative approach for clean code.
- Program structure should be decomposed into clear steps: prompt, input reading, condition checking, and output printing.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Challenges :While Loop

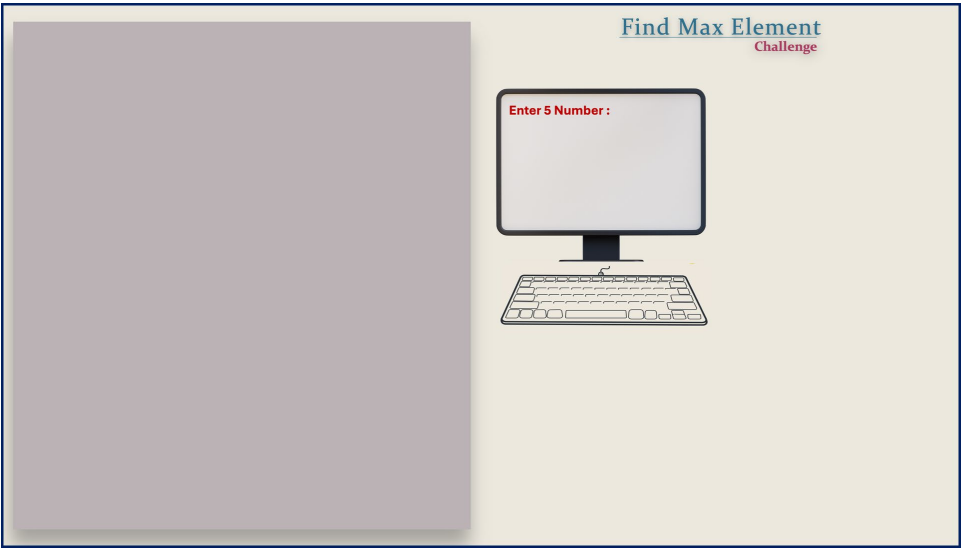
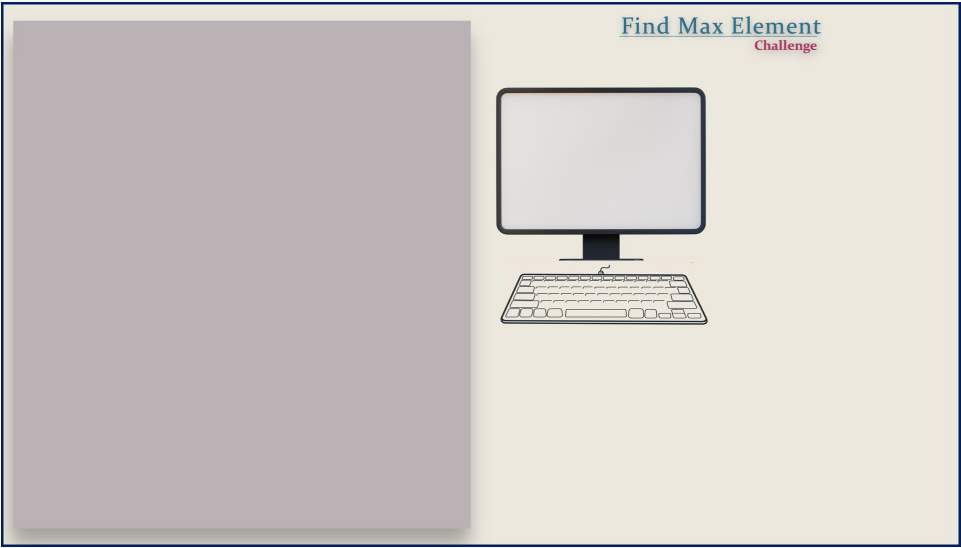
Challenges :While Loop

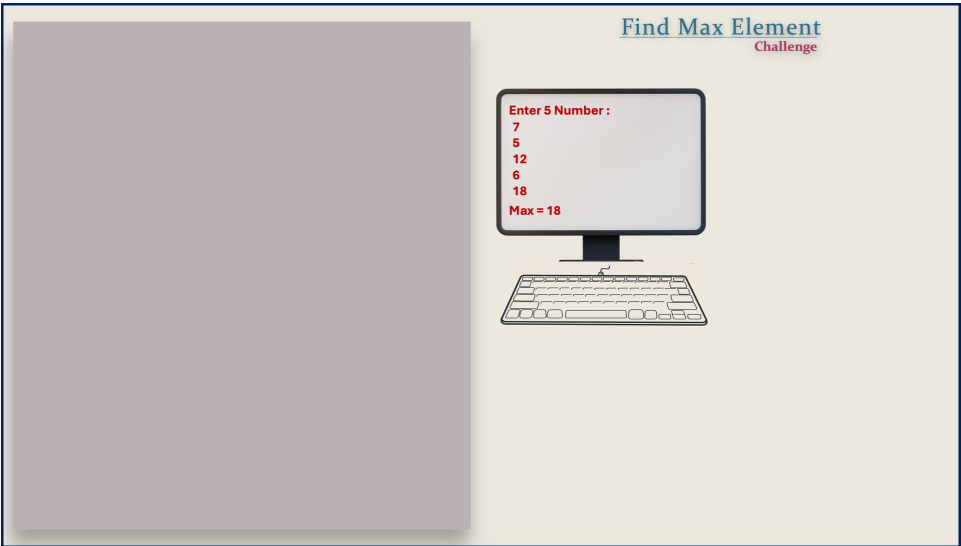
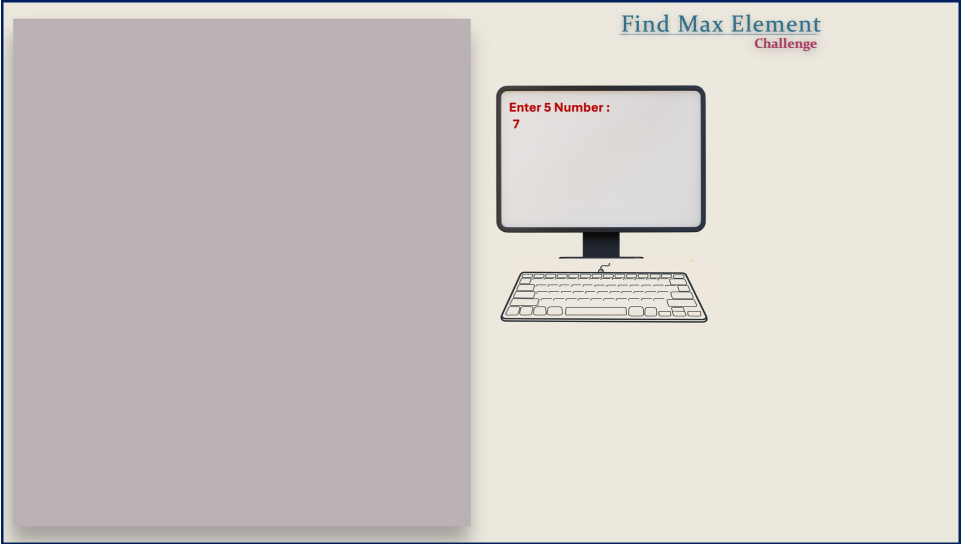
- Find Max Element

Challenges :While Loop

- Find Max Element
- Find Max Min Element

Find Max Element Challenge





Find Max Element

Challenge

Enter 5 Number :

Max

0

Find Max Element

Challenge

Enter 5 Number :

7

x

Max

0

Find Max Element

Challenge

Enter 5 Number :

7

x

7

Max

0

if x > Max :

Max = x

Find Max Element

Challenge

Enter 5 Number :

7

5

12

x

12

Max

7

if x > Max :

Max = x

Find Max Element

Challenge

Enter 5 Number :

7

5

12

6

18

x

18

Max

18

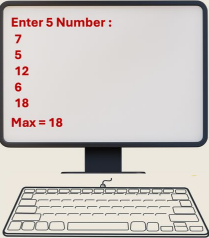
if x > Max :

Max = x

Find Max Element

Challenge

Enter 5 Number :
7
5
12
6
18
Max = 18

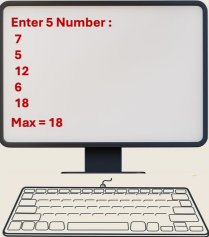


- Number of Elements ?

Find Max Element

Challenge

Enter 5 Number :
7
5
12
6
18
Max = 18



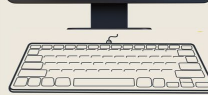
- Number of Elements ?
- Counter for n Elements

Find Max Element

Challenge

Enter 5 Number :

7
5
12
6
18
Max = 18



- Number of Elements ?
- Counter for n Elements
- Read n Elements

Find Max Element

Challenge

Enter 5 Number :

7
5
12
6
18
Max = 18



- Number of Elements ?
- Counter for n Elements
- Read n Elements
- Find Max of n Elements

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- Read n Elements
- Find Max of n Elements

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- Read n Elements
- Find Max of n Elements

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- Read n Elements
- Find Max of n Elements

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- Read n Elements
- Find Max of n Elements

n = 5

Find Max Element

Challenge

- ➡ • Number of Elements ?
- Counter for n Elements
- Read n Elements
- Find Max of n Elements

```
n = 5  
print( 'Enter ' , n, 'Numbers' )
```

Find Max Element

Challenge

- Number of Elements ?
- »»» • Counter for n Elements
- Read n Elements
- Find Max of n Elements

```
n = 5  
print( 'Enter ', n, 'Numbers' )
```

Find Max Element

Challenge

- Number of Elements ?
- »»» • Counter for n Elements
- Read n Elements
- Find Max of n Elements

```
n = 5  
print( 'Enter ', n, 'Numbers' )  
i = 0
```

Find Max Element Challenge

- Number of Elements ?
- »»» • Counter for n Elements
- Read n Elements
- Find Max of n Elements

```
n = 5
print( 'Enter ', n, 'Numbers' )
i = 0

while i < n :
```

Find Max Element Challenge

- Number of Elements ?
- »»» • Counter for n Elements
- Read n Elements
- Find Max of n Elements

```
n = 5
print( 'Enter ', n, 'Numbers' )
i = 0

while i < n :
    i = i + 1
```

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- »»» • Read n Elements
- Find Max of n Elements

```
n = 5
print( 'Enter ', n, 'Numbers' )
i = 0

while i < n :
    i = i + 1
```

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- »»» • Read n Elements
- Find Max of n Elements

```
n = 5
print( 'Enter ', n, 'Numbers' )
i = 0

while i < n :
    i = i + 1
    x = int(input( " "))
```

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- Read n Elements
- ➡➡➡ • Find Max of n Elements

```
n = 5
print( 'Enter ', n, 'Numbers' )
i = 0

while i < n :
    i = i + 1
    x = int(input( " "))
```

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- Read n Elements
- ➡➡➡ • Find Max of n Elements

```
n = 5
print( 'Enter ', n, 'Numbers' )
i = 0
Max = 0
while i < n :
    i = i + 1
    x = int(input( " "))
```

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- Read n Elements
- ➡➡➡ • Find Max of n Elements

```
n = 5
print( 'Enter ', n, 'Numbers' )
i = 0
Max = 0
while i < n :
    i = i + 1
    x = int(input( " "))
    if x > Max :
```

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- Read n Elements
- ➡➡➡ • Find Max of n Elements

```
n = 5
print( 'Enter ', n, 'Numbers' )
i = 0
Max = 0
while i < n :
    i = i + 1
    x = int(input( " "))
    if x > Max :
        Max = x
```

Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- Read n Elements
- Find Max of n Elements

```
n = 5
print( 'Enter ', n, 'Numbers' )
i = 0
Max = 0
while i < n :
    i = i + 1
    x = int(input( " "))
    if x > Max :
        Max = x
```


Find Max Element

Challenge

- Number of Elements ?
- Counter for n Elements
- Read n Elements
- Find Max of n Elements

```
n = 5
print( 'Enter ', n, 'Numbers' )
i = 0
Max = 0
while i < n :
    i = i + 1
    x = int(input( " "))
    if x > Max :
        Max = x
print( 'Max Element: ', Max )
```

- Find Max & Min Element

Find Max & Min Element

Challenge

Find Max & Min Element

Challenge

Enter 5 Number :

7

5

12

6

18

Max = 18

Find Max & Min Element

Challenge

Enter 5 Number :

7

5

12

6

18

Max = 18

Min = 5

Find Max & Min Element

Challenge

```
n = 5
print( 'Enter', n, 'Numbers' )
i = 0
Max = 0
while i < n :
    i = i + 1
    x = int(input( " "))
    if x > Max :
        Max = x
print( 'Max Element: ', Max )
```

Find Max & Min Element

Challenge

```
if x > Max :
    Max = x
```

Find Max & Min Element

Challenge

```
if x > Max :  
    Max = x  
else:  
    Min = x
```

Find Max & Min Element

Challenge

Max
12

```
if x > Max :  
    Max = x  
else:  
    Min = x
```

Find Max & Min Element

Challenge

Max

12

Min

5

```
if x > Max :  
    Max = x  
else:  
    Min = x
```

Find Max & Min Element

Challenge

x

8

Max

12

Min

5

if x > Max :

Max = x

else:

Min = x

Find Max & Min Element

Challenge

x

8

Max

12

Min

5

if x > Max :

Max = x

if

Min = x

Find Max & Min Element

Challenge

x

8

Max

12

Min

5

if x > Max :
 Max = x

if x < Min :
 Min = x

Find Max & Min Element

Challenge

if x > Max :
 Max = x

if x < Min :
 Min = x

Find Max & Min Element

Challenge

```
if x > Max :  
    Max = x  
if x < Min :  
    Min = x
```

Find Max & Min Element

Challenge

Max =

```
if x > Max :  
    Max = x  
if x < Min :  
    Min = x
```

Find Max & Min Element

Challenge

```
Max = float( '-inf' )
```

```
if x > Max :  
    Max = x
```

```
if x < Min :  
    Min = x
```

Find Max & Min Element

Challenge

```
Max = float( '-inf' )
```

```
Min =
```

```
if x > Max :  
    Max = x
```

```
if x < Min :  
    Min = x
```

Find Max & Min Element

Challenge

```
Max = float( '-inf' )
```

```
Min = float( 'inf' )
```

```
if x > Max :
```

```
    Max = x
```

```
if x < Min :
```

```
    Min = x
```